1.37 2D Matrices: Rotations

Worked Solutions

1 a)

Rotation, anticlockwise about the origin, by 90°.

Alternative Method:

Rotation, clockwise about the origin, by 270°.

1 b)

Rotation, about the origin, by 180°.

1 c)

Rotation, anticlockwise about the origin, by 270°.

Alternative Method:

Rotation, clockwise about the origin, by 90°.

2 a)

$$\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

2 b)

$$\mathbf{A}^2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}.$$

2 c)

$$\mathbf{A}^3 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^3 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

2 d)

$$\mathbf{A}^4 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^4 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

2 e)

$$\mathbf{A}^5 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^5 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

2 f)

$$\mathbf{A}^{2025} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^{2025} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

3)

This is a rotation, anticlockwise about the origin, by 90°, so area is preserved. Hence the area of the shape's image is 4 units^2.

4)

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 6 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \end{pmatrix}.$$

So the coordinates of B are $(2, 6)$.

5)

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \sqrt{2} \\ -2\sqrt{2} \end{pmatrix} = \begin{pmatrix} -\sqrt{2} \\ 2\sqrt{2} \end{pmatrix}.$$

So the coordinates of B are $(-\sqrt{2}, 2\sqrt{2})$.

6)

$$\frac{990^\circ}{90^\circ} = 11 \qquad 11 \div 4 = 2 \text{ r } 3$$

A rotation, anticlockwise about the origin, by 90° can be represented by the matrix:

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \quad \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^3 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^4 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^5 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

...

So $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^{11} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^3 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 5 \\ -17 \end{pmatrix} = \begin{pmatrix} -17 \\ -5 \end{pmatrix}$$

So the coordinates of B are $(-17, -5)$.

7)

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 & 6 \\ 3 & 8 & -1 \end{pmatrix} = \begin{pmatrix} 3 & 8 & -1 \\ -1 & -2 & -6 \end{pmatrix}$$

So the coordinates of the vertices of T' are $(3, -1)$, $(8, -2)$ and $(-1, -6)$.

8)

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 10 & 10 \\ 2 & 6 & 6 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 & -10 & -10 \\ -2 & -6 & -6 & -2 \end{pmatrix}$$

So the coordinates of the vertices of R' are $(0, -2)$, $(0, -6)$, $(-10, -6)$ and $(-10, -2)$.

9 a)

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ -5 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$$

A rotation, anticlockwise about the origin, by 90°.

9 b)

$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

9 c)

Distance from $(2, -5)$ to the origin is $\sqrt{2^2 + (-5)^2} = \sqrt{29}$.

Length of arc $= \frac{90}{360} \times 2\pi \times \sqrt{29} = \frac{\pi\sqrt{29}}{2}$ units.

10)

The transformation appears to be a rotation, clockwise about the origin, by 90° (or a rotation, anticlockwise about the origin, by 270°):

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} -2 & -5 & -5 & -2 \\ 2 & 4 & 6 & 4 \end{pmatrix} = \begin{pmatrix} a & b & c & 4 \\ d & e & f & 2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2 & 4 & 6 & 4 \\ 2 & 5 & 5 & 2 \end{pmatrix} = \begin{pmatrix} a & b & c & 4 \\ d & e & f & 2 \end{pmatrix}$$

So the coordinates of the vertices of the image are $(2, 2)$, $(4, 5)$, $(6, 5)$ and $(4, 2)$.

$\therefore \mathbf{M} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and $A'(2, 2)$, $B'(4, 5)$, $C'(6, 5)$.

11)

$\overrightarrow{BA} = \overrightarrow{OA} - \overrightarrow{OB} = \begin{pmatrix} 9 \\ 4 \end{pmatrix} - \begin{pmatrix} 6 \\ -3 \end{pmatrix} = \begin{pmatrix} 3 \\ 7 \end{pmatrix}$.

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 3 \\ 7 \end{pmatrix} = \begin{pmatrix} -7 \\ 3 \end{pmatrix}$$

$\overrightarrow{OA'} = \overrightarrow{OB} + \begin{pmatrix} -7 \\ 3 \end{pmatrix} = \begin{pmatrix} 6 \\ -3 \end{pmatrix} + \begin{pmatrix} -7 \\ 3 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$.

So the coordinates of A' are $(-1, 0)$.

12 a)

$$\begin{pmatrix} \cos 45° & -\sin 45° \\ \sin 45° & \cos 45° \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

12 b)

A rotation by 45°, clockwise about the origin is equivalent to a rotation by 315°, anticlockwise about the origin.

$$\begin{pmatrix} \cos 315° & -\sin 315° \\ \sin 315° & \cos 315° \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

12 c)

$$\begin{pmatrix} \cos 135^\circ & -\sin 135^\circ \\ \sin 135^\circ & \cos 135^\circ \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

12 d)

A rotation by 135°, clockwise about the origin is equivalent to a rotation by 225°, anticlockwise about the origin.

$$\begin{pmatrix} \cos 225^\circ & -\sin 225^\circ \\ \sin 225^\circ & \cos 225^\circ \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

13)

$$\begin{pmatrix} \cos 45^\circ & -\sin 45^\circ \\ \sin 45^\circ & \cos 45^\circ \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \begin{pmatrix} -2 & 8 & 4 \\ 5 & 3 & -2 \end{pmatrix} = \begin{pmatrix} -\frac{7\sqrt{2}}{2} & \frac{5\sqrt{2}}{2} & 3\sqrt{2} \\ \frac{3\sqrt{2}}{2} & \frac{11\sqrt{2}}{2} & \sqrt{2} \end{pmatrix}$$

So the coordinates of the vertices of T' are $(-\frac{7\sqrt{2}}{2}, \frac{3\sqrt{2}}{2})$, $(\frac{5\sqrt{2}}{2}, \frac{11\sqrt{2}}{2})$ and $(3\sqrt{2}, \sqrt{2})$.

14)

$360^\circ - 225^\circ = 135^\circ$.

$$\begin{pmatrix} \cos 135^\circ & -\sin 135^\circ \\ \sin 135^\circ & \cos 135^\circ \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

$$\begin{pmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix} \begin{pmatrix} -3 & -3 & 6 & 6 \\ 2 & 4 & 4 & 2 \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & -5\sqrt{2} & -4\sqrt{2} \\ -\frac{5\sqrt{2}}{2} & -\frac{7\sqrt{2}}{2} & \sqrt{2} & 2\sqrt{2} \end{pmatrix}$$

So the coordinates of the vertices of R' are $(\frac{\sqrt{2}}{2}, -\frac{5\sqrt{2}}{2})$, $(-\frac{\sqrt{2}}{2}, -\frac{7\sqrt{2}}{2})$, $(-5\sqrt{2}, \sqrt{2})$ and $(-4\sqrt{2}, 2\sqrt{2})$.

15)

$$\begin{pmatrix} \cos 135^\circ & -\sin 135^\circ \\ \sin 135^\circ & \cos 135^\circ \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

$$\begin{pmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix} \begin{pmatrix} -\sqrt{2} & k\sqrt{2} & 1 \\ 3\sqrt{2} & 5\sqrt{2} & -1 \end{pmatrix} = \begin{pmatrix} -2 & -13 & 0 \\ -4 & 3 & \sqrt{2} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -2 & -k-5 & 0 \\ -4 & k-5 & \sqrt{2} \end{pmatrix} = \begin{pmatrix} -2 & -13 & 0 \\ -4 & 3 & \sqrt{2} \end{pmatrix}$$

$\therefore k - 5 = 3 \Rightarrow k = 8$.

16 a)

$$\begin{pmatrix} \cos 30^\circ & -\sin 30^\circ \\ \sin 30^\circ & \cos 30^\circ \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$$

16 b)

$$\begin{pmatrix} \cos 60^\circ & -\sin 60^\circ \\ \sin 60^\circ & \cos 60^\circ \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

16 c)

$$\begin{pmatrix} \cos 120^\circ & -\sin 120^\circ \\ \sin 120^\circ & \cos 120^\circ \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$$

16 d)

$$\begin{pmatrix} \cos 150^\circ & -\sin 150^\circ \\ \sin 150^\circ & \cos 150^\circ \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix}$$

16 e)

A rotation by 210°, clockwise about the origin, is the same as a rotation by $360^\circ - 210^\circ = 150^\circ$, anticlockwise about the origin:

$$\begin{pmatrix} \cos 150^\circ & -\sin 150^\circ \\ \sin 150^\circ & \cos 150^\circ \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix}$$

17)

$$\begin{pmatrix} \cos 18^\circ & -\sin 18^\circ \\ \sin 18^\circ & \cos 18^\circ \end{pmatrix} = \begin{pmatrix} 0.951 & -0.309 \\ 0.309 & 0.951 \end{pmatrix} \quad (3 \text{ SF})$$

18)

$360^\circ - 25^\circ = 335^\circ$.

$$\begin{pmatrix} \cos 335^\circ & -\sin 335^\circ \\ \sin 335^\circ & \cos 335^\circ \end{pmatrix} = \begin{pmatrix} 0.906 & 0.423 \\ -0.423 & 0.906 \end{pmatrix} \quad (3 \text{ SF})$$

19 a)

$\cos\theta = -\frac{1}{2} \Rightarrow \theta = \{120^\circ, 240^\circ\}$

$\sin\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = \{60^\circ, 120^\circ\}$

Rotation, anticlockwise about the origin, by 120°.

19 b)

$\cos\theta = -\frac{\sqrt{3}}{2} \Rightarrow \theta = \{150^\circ, 210^\circ\}$

$\sin\theta = -\frac{1}{2} \Rightarrow \theta = \{210^\circ, 330^\circ\}$

Rotation, anticlockwise about the origin, by 210°.

19 c)

$\cos\theta = -\frac{\sqrt{2}}{2} \Rightarrow \theta = \{135°, 225°\}$

$\sin\theta = \frac{\sqrt{2}}{2} \Rightarrow \theta = \{45°, 135°\}$

Rotation, anticlockwise about the origin, by 135°.

19 d)

$\cos\theta = \frac{\sqrt{10+2\sqrt{5}}}{4} \Rightarrow \theta = \{18°, 342°\}$

$\sin\theta = \frac{-1+\sqrt{5}}{4} \Rightarrow \theta = \{18°, 162°\}$

Rotation, anticlockwise about the origin, by 18°.

20)

$\cos\theta = \frac{5}{\sqrt{41}} \Rightarrow \theta = \{38.7°, 321°\}$ (3 SF)

$\sin\theta = \frac{4}{\sqrt{41}} \Rightarrow \theta = \{38.7°, 141°\}$ (3 SF)

So $\theta = 38.7°$ (3 SF).

21)

$\cos\theta = \frac{\sqrt{6}-\sqrt{2}}{4} \Rightarrow \theta = \{75°, 285°\}$

$\sin\theta = -\frac{\sqrt{6}+\sqrt{2}}{4} \Rightarrow \theta = \{255°, 285°\}$

So $\theta = 285°$.

22 a)

$\cos\theta = \frac{13}{85} \Rightarrow \theta = \{81.2°, 279°\}$ (3 SF)

$\sin\theta = \frac{84}{85} \Rightarrow \theta = \{81.2°, 98.8°\}$ (3 SF)

So yes, this represents a rotation, anticlockwise about the origin, by 81.2° (3 SF).

22 b)

$\cos\theta = -\frac{13}{85} \Rightarrow \theta = \{98.8°, 261°\}$ (3 SF)

$\cos\theta = \frac{13}{85} \Rightarrow \theta = \{81.2°, 279°\}$ (3 SF)

$\sin\theta = \frac{84}{85} \Rightarrow \theta = \{81.2°, 98.8°\}$ (3 SF)

So no, this does not represent a rotation as there is no common angle in the solutions to the three equations.

22 c)

$\cos\theta = \frac{13}{85} \Rightarrow \theta = \{81.2°, 279°\}$ (3 SF)

$\sin\theta = -\frac{84}{85} \Rightarrow \theta = \{261°, 279°\}$ (3 SF)

So yes, this represents a rotation, anticlockwise about the origin, by 279° (3 SF).

22 d)

$\cos\theta = \frac{84}{85} \Rightarrow \theta = \{8.80°, 351°\}$ (3 SF)

$\sin\theta = \frac{13}{85} \Rightarrow \theta = \{8.80°, 171°\}$ (3 SF)

So yes, this represents a rotation, anticlockwise about the origin, by 8.80° (3 SF).

23)

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}\begin{pmatrix} 2 \\ 10 \end{pmatrix} = \begin{pmatrix} -10 \\ -2 \end{pmatrix} \Rightarrow \begin{pmatrix} 2a-10b \\ 2b+10a \end{pmatrix} = \begin{pmatrix} -10 \\ -2 \end{pmatrix}$$

$2a - 10b = -10 \quad (1)$

$10a + 2b = -2 \quad (2)$

$\therefore a = -\frac{5}{13}$ and $b = \frac{12}{13}$.

$\cos\theta = -\frac{5}{13} \Rightarrow \theta = \{113°, 247°\}$ (3 SF)

$\sin\theta = \frac{12}{13} \Rightarrow \theta = \{67.4°, 113°\}$ (3 SF)

So $\theta = 113°$ to 3 SF.

24)

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}\begin{pmatrix} 15 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ -15 \end{pmatrix} \Rightarrow \begin{pmatrix} 15a+2b \\ 15b-2a \end{pmatrix} = \begin{pmatrix} 2 \\ -15 \end{pmatrix}$$

$15a + 2b = 2 \quad (1)$

$-2a + 15b = -15 \quad (2)$

$\therefore a = \frac{60}{229}$ and $b = -\frac{221}{229}$.

$\cos\theta = \frac{60}{229} \Rightarrow \theta = \{74.8°, 285°\}$ (3 SF)

$\sin\theta = -\frac{221}{229} \Rightarrow \theta = \{255°, 285°\}$ (3 SF)

So $\theta = 285°$ to 3 SF.

25)

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} 1 \\ 5 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ \frac{\sqrt{103}}{2} \end{pmatrix}$$

$\cos\theta - 5\sin\theta = \frac{1}{2} \Rightarrow \cos\theta = \frac{1}{2} + 5\sin\theta \quad (1)$

$\sin\theta + 5\cos\theta = \frac{\sqrt{103}}{2} \quad (2)$

Substitute (1) into (2):

$$\sin\theta + 5\left(\frac{1}{2} + 5\sin\theta\right) = \frac{\sqrt{103}}{2}$$

$$\Rightarrow \sin\theta + \frac{5}{2} + 25\sin\theta = \frac{\sqrt{103}}{2}$$

$$\Rightarrow 26\sin\theta = \frac{\sqrt{103}}{2} - \frac{5}{2}$$

$$\Rightarrow \sin\theta = \frac{\sqrt{103}}{52} - \frac{5}{52} \Rightarrow \theta = \{5.68°, 174°\} \quad (3 \text{ SF})$$

$$\cos\theta = \frac{1}{2} + 5\left(\frac{\sqrt{103}}{52} - \frac{5}{52}\right) = \frac{1}{52} + \frac{5\sqrt{103}}{52} \Rightarrow \theta = \{5.68°, 354°\} \quad (3 \text{ SF})$$

$$\mathbf{M} = \begin{pmatrix} \frac{1}{52} + \frac{5\sqrt{103}}{52} & \frac{5}{52} - \frac{\sqrt{103}}{52} \\ \frac{\sqrt{103}}{52} - \frac{5}{52} & \frac{1}{52} + \frac{5\sqrt{103}}{52} \end{pmatrix} = \begin{pmatrix} 0.995 & -0.099 \\ 0.099 & 0.995 \end{pmatrix} \quad (3 \text{ DP})$$

$$\begin{pmatrix} \frac{1}{52} + \frac{5\sqrt{103}}{52} & \frac{5}{52} - \frac{\sqrt{103}}{52} \\ \frac{\sqrt{103}}{52} - \frac{5}{52} & \frac{1}{52} + \frac{5\sqrt{103}}{52} \end{pmatrix}\begin{pmatrix} 3 & 4 & 1 \\ 1 & 2 & 5 \end{pmatrix} = \begin{pmatrix} \frac{4+7\sqrt{103}}{26} & \frac{7+9\sqrt{103}}{26} & \frac{1}{2} \\ \frac{-7+4\sqrt{103}}{26} & \frac{-9+7\sqrt{103}}{26} & \frac{\sqrt{103}}{2} \end{pmatrix}$$

So $A'\left(\frac{4+7\sqrt{103}}{26}, \frac{-7+4\sqrt{103}}{26}\right) = A'(2.886, 1.292)$ (3 DP).

$B'\left(\frac{7+9\sqrt{103}}{26}, \frac{-9+7\sqrt{103}}{26}\right) = B'(3.782, 2.386)$ (3 DP).

With $\mathbf{M} = \begin{pmatrix} \frac{1}{52}+\frac{5\sqrt{103}}{52} & \frac{5}{52}-\frac{\sqrt{103}}{52} \\ \frac{\sqrt{103}}{52}-\frac{5}{52} & \frac{1}{52}+\frac{5\sqrt{103}}{52} \end{pmatrix} = \begin{pmatrix} 0.995 & -0.099 \\ 0.099 & 0.995 \end{pmatrix}$ (3 DP).

26)

$\overrightarrow{BA} = \overrightarrow{OA} - \overrightarrow{OB} = \begin{pmatrix} -5 \\ 1 \end{pmatrix} - \begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} -7 \\ 1 \end{pmatrix}$.

$$\begin{pmatrix} \cos 60° & -\sin 60° \\ \sin 60° & \cos 60° \end{pmatrix} \begin{pmatrix} -7 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} -7 \\ 1 \end{pmatrix} = \begin{pmatrix} -\frac{7}{2}-\frac{\sqrt{3}}{2} \\ \frac{1}{2}-\frac{7\sqrt{3}}{2} \end{pmatrix}$$

$\overrightarrow{OA'} = \overrightarrow{OB} + \begin{pmatrix} -\frac{7}{2}-\frac{\sqrt{3}}{2} \\ \frac{1}{2}-\frac{7\sqrt{3}}{2} \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix} + \begin{pmatrix} -\frac{7}{2}-\frac{\sqrt{3}}{2} \\ \frac{1}{2}-\frac{7\sqrt{3}}{2} \end{pmatrix} = \begin{pmatrix} -\frac{3}{2}-\frac{\sqrt{3}}{2} \\ \frac{1}{2}-\frac{7\sqrt{3}}{2} \end{pmatrix}$.

So the coordinates of A' are $(-\frac{3}{2}-\frac{\sqrt{3}}{2}, \frac{1}{2}-\frac{7\sqrt{3}}{2})$.

27)

Let P have coordinates (a, b):

$$\overrightarrow{PA} = \overrightarrow{OA} - \overrightarrow{OP} = \begin{pmatrix} 0 \\ 8 \end{pmatrix} - \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -a \\ 8-b \end{pmatrix}$$

$$\begin{pmatrix} \cos 30° & -\sin 30° \\ \sin 30° & \cos 30° \end{pmatrix} \begin{pmatrix} -a \\ 8-b \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix} \begin{pmatrix} -a \\ 8-b \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{\sqrt{3}}{2}a - 4 + \frac{1}{2}b \\ -\frac{1}{2}a + 4\sqrt{3} - \frac{\sqrt{3}}{2}b \end{pmatrix}$$

$\overrightarrow{OA'} = \overrightarrow{OP} + \begin{pmatrix} -\frac{\sqrt{3}}{2}a - 4 + \frac{1}{2}b \\ -\frac{1}{2}a + 4\sqrt{3} - \frac{\sqrt{3}}{2}b \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} -\frac{\sqrt{3}}{2}a - 4 + \frac{1}{2}b \\ -\frac{1}{2}a + 4\sqrt{3} - \frac{\sqrt{3}}{2}b \end{pmatrix}$

$$= \begin{pmatrix} (1-\frac{\sqrt{3}}{2})a - 4 + \frac{1}{2}b \\ -\frac{1}{2}a + 4\sqrt{3} + (1-\frac{\sqrt{3}}{2})b \end{pmatrix} = \begin{pmatrix} -5+\sqrt{3} \\ 3+3\sqrt{3} \end{pmatrix}$$

$(1-\frac{\sqrt{3}}{2})a - 4 + \frac{1}{2}b = -5+\sqrt{3} \Rightarrow (2-\sqrt{3})a - 8 + b = -10 + 2\sqrt{3}$

$$\Rightarrow (2-\sqrt{3})a + b = -2 + 2\sqrt{3} \quad (1)$$

$-\frac{1}{2}a + 4\sqrt{3} + (1-\frac{\sqrt{3}}{2})b = 3 + 3\sqrt{3} \Rightarrow -a + 8\sqrt{3} + (2-\sqrt{3})b = 6 + 6\sqrt{3}$

$$\Rightarrow -a + (2-\sqrt{3})b = 6 - 2\sqrt{3} \quad (2)$$

$\therefore a = -2$ and $b = 2$.

So the coordinates of P are $(-2, 2)$.